Semantic Concurrency Limits in Large Language Models

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High-dimensional embedding spaces can host many semantic directions with small mutual overlap. But small overlaps are not zero: when many directions are jointly active, their residual interference accumulates and limits what a finite readout channel can recover. We formulate this as a distinction between *kinetic capacity*—what the geometry can host—and *epistemic accessibility*—what readout can recover. The two sides are summarized by $N \lesssim \exp(c\, d_{\text{eff}}\, \varepsilon^2)$ for coexistence and $\sigma_{\text{int}} \sim \sqrt{k/d_{\text{eff}}}$ for simultaneous readout. Thus dimension acts not merely as storage capacity but as semantic concurrency bandwidth. On this geometric foundation we propose a separate hypothesis: some polysemous tokens may be organized around stable token-associated hinge directions, with sense information carried by low-dimensional subspaces in the hinge-perpendicular carrier. The capacity/accessibility distinction is the main claim; the hinge hypothesis is a stronger, separately falsifiable empirical proposal.

I. INTRODUCTION

A finite-dimensional activation space faces a simple tension. High dimension lets many semantic features coexist as nearly orthogonal directions. Finite readout prevents too many of them from being simultaneously recovered from one channel. This paper makes that tension quantitative.

The two sides obey different scalings. On the capacity side, concentration of measure permits exponentially many candidate semantic directions with pairwise overlap at most ε,

$$N \lesssim \exp(c\, d_{\text{eff}}\, \varepsilon^2). \tag{1}$$

On the readout side, the same small residual overlaps add in quadrature over an active set of size k, giving an interference scale

$$\sigma_{\text{int}} \sim \sqrt{k/d_{\text{eff}}}. \tag{2}$$

The first relation is an existential packing estimate. The second is a typical-case readout-noise estimate. Together they say that dimension is a *semantic concurrency bandwidth*: it does not mainly limit how many meanings can be geometrically stored, but how many independently recoverable semantic components can be jointly active.

Polysemy is the cleanest example. A single lexical item, such as "bank", can access several mutually exclusive meanings, only one of which is selected in a given context. The same tension appears more generally in feature superposition, role binding, and compositional semantics [1]: many latent semantic degrees of freedom must coexist, while only a context-selected subset is read out. We use polysemy as a controlled entry point because it makes contextual selection explicit. In lexical semantics, polysemy is distinguished from homonymy by the relatedness of senses [2]. Modern Transformer-based LLMs implement contextual selection through dynamic contextual embeddings: a token representation is repeatedly transformed by self-attention and MLP layers as it absorbs information from its sentential environment [3].

The paper makes two claims. The first is the geometric capacity claim: concentration permits large semantic capacity, while accumulated interference limits simultaneous accessibility. This claim is architecture-independent. The second is a stronger organizational hypothesis: for some canonically polysemous tokens, readout may be structured around a stable token-associated *hinge* direction, with sense information carried by low-dimensional subspaces in its orthogonal complement. The hinge hypothesis is not implied by concentration geometry. It is an empirical proposal, and Sec. V D states how it can fail.

A serious caveat is effective dimension. Trained representations are often anisotropic: they occupy a narrow cone rather than the full ambient space. The dimension controlling concentration is therefore not necessarily the raw embedding dimension d, but an effective dimension d_{eff}. If d_{eff} is small, the capacity claim weakens accordingly. For example, at $d_{\text{eff}} \sim 50$ and $\varepsilon = 0.1$, the exponent $c d_{\text{eff}} \varepsilon^2$ is only a small constant. The framework therefore applies to the approximately isotropic geometry actually used by the representation, possibly after whitening or restriction to a participation subspace. Determining which layers, models, and preprocessing regimes admit a meaningful d_{eff} is an empirical priority.

The hinge hypothesis has a narrower domain. It is meant for canonically polysemous tokens with established sense inventories, such as WordNet-, OntoNotes-, SemCor-, or WiC-style sense distinctions [4] with enough contextual instances for testing. Failure on a representative sample, against the controls in Sec. V D, would count against hinge anchoring even if the broader capacity/accessibility picture survives.

The proposal is adjacent to the superposition hypothesis in mechanistic interpretability [5], according to which neural networks represent more features than they have

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neurons by packing features into nearly orthogonal directions. The present paper separates the geometric background from a possible semantic organization within it. The general superposition picture concerns an overcomplete dictionary of feature directions and sparse active sets. The hinge picture adds the more specific possibility that some contextual readout is organized around stable token-associated directions.

Section II develops the capacity framework: quasi-orthogonality, Lévy's lemma, exponential packing, and the kinetic/epistemic distinction. Section III introduces the hinge-and-frame hypothesis and its relation to dynamic contextual embeddings. Section IV connects the picture to attention as soft routing rather than literal frame reconstruction. Section V discusses superposition, sparse autoencoders, limitations, empirical tests, and the limited quantum analogy.

a. Notation. Throughout, $\mathbb{R}^d$ is Euclidean d-space with inner product $u^\top v$ and norm $\|v\| = \sqrt{v^\top v}$; $S^{d-1} = \{v \in \mathbb{R}^d : \|v\| = 1\}$ is the unit sphere; I is the identity matrix; $\|\cdot\|_{\mathrm{op}}$ is the operator norm; $A \succeq 0$ means positive semidefinite; and $\mathbb{P}$ and $\mathbb{E}$ denote probability and expectation under the uniform measure on the relevant sphere unless stated otherwise. Universal positive constants are denoted c, c', possibly changing from line to line. The symbol d_{eff} denotes the effective dimension controlling concentration in the representation being probed.

II. CONCENTRATION GEOMETRY AND CAPACITY

This section is architecture-independent. It gives the geometric capacity constraint on which the later hinge hypothesis builds.

A. Quasi-orthogonality

Let $v_1, \ldots, v_N \in \mathbb{R}^d$ be unit vectors. Strict orthogonality, $v_i^\top v_j = 0$ for $i \neq j$, allows at most $N = d$ such vectors. Semantic encoding does not require exact orthogonality. It requires approximate distinguishability. We therefore call a family ε-quasi-orthogonal if

$$|v_i^\top v_j| \leq \varepsilon, \qquad i \neq j. \tag{3}$$

This is the relevant notion for high-dimensional feature storage: small overlap is tolerable, provided readout can absorb the resulting interference.

B. Lévy's lemma and linear overlaps

High-dimensional spheres make quasi-orthogonality generic. Lévy's lemma states that if $f : S^{d-1} \to \mathbb{R}$ is L-Lipschitz, then

$$\mathbb{P}\big(|f(v) - \mathbb{E}f| \geq \delta\big) \leq 2\exp\left(-\frac{c\,d\,\delta^2}{L^2}\right), \tag{4}$$

for a universal $c > 0$ [6, 7]. Apply this to the linear functional $f_u(v) = u^\top v$. It is 1-Lipschitz and has mean zero by symmetry. Hence

$$\mathbb{P}\big(|u^\top v| \geq \varepsilon\big) \leq 2\exp(-c\,d\,\varepsilon^2). \tag{5}$$

For normalized surface measure one obtains the explicit form $2\exp(-(d-1)\varepsilon^2/2)$ [6]. Thus two random unit vectors are nearly orthogonal with high probability, and typical overlap amplitudes scale as $d^{-1/2}$.

In trained models, d should be replaced by d_{eff}, the dimension actually controlling concentration. Anisotropy, low intrinsic dimension, or a narrow activation cone reduce d_{eff}. Whitening or projection to a participation subspace may increase the effective dimension relevant for the analysis.

a. Relation to Johnson–Lindenstrauss. The Johnson–Lindenstrauss lemma preserves pairwise distances among n points after projection to dimension $k = O(\varepsilon^{-2}\log n)$ [8]. It is driven by the same concentration phenomenon as Eq. (5); indeed JL can be derived from sphere concentration applied to projection norms [9]. Here Lévy's lemma is the more direct tool: the issue is not projecting a finite dataset, but the typical overlap between semantic directions in the representation space.

C. Exponential packing

Sample N unit vectors independently and uniformly from S^{d-1}. By Eq. (5) and a union bound over pairs,

$$\mathbb{P}\big(\exists\, i \neq j : \ |v_i^\top v_j| \geq \varepsilon\big) < N^2 \exp(-c\,d_{\mathrm{eff}}\,\varepsilon^2). \tag{6}$$

This probability is below one, and hence an ε-quasi-orthogonal family exists, provided

$$N < \exp\left(\frac{c}{2}\,d_{\mathrm{eff}}\,\varepsilon^2\right). \tag{7}$$

This is the kinetic capacity estimate: in high effective dimension, many candidate semantic directions can coexist with small pairwise overlap.

The estimate controls pairwise overlap only. It does not guarantee that a large frame is well-conditioned. By Gershgorin, $|e_i^\top e_j| \leq \varepsilon$ gives only $\|G - I\|_{\mathrm{op}} \leq (n-1)\varepsilon$ for an n-element Gram matrix, which is useless when n is large. Frame-level readout therefore needs a separate conditioning assumption, introduced in Sec. III B.

D. Kinetic capacity versus epistemic accessibility

The packing bound says what can coexist. It does not say what can be read. We call the coexistence supply *kinetic semantic capacity*. We call the recoverable subset *epistemic semantic accessibility*. The distinction is the central point of the paper.

Let N candidate features be represented by unit directions $w_1, \ldots, w_N \in \mathbb{R}^{d_{\text{eff}}}$. In a given context, suppose an active set S of size k contributes amplitudes x_j, producing

$$h = \sum_{j \in S} x_j w_j. \tag{8}$$

A linear readout of feature i gives

$$w_i^{\mathsf{T}} h = x_i + \sum_{\substack{j \in S \\ j \neq i}} x_j \, w_i^{\mathsf{T}} w_j. \tag{9}$$

The first term is signal; the second is interference. For random incoherent directions,

$$\mathbb{E}[w_i^{\mathsf{T}} w_j] = 0, \qquad \mathrm{Var}(w_i^{\mathsf{T}} w_j) \simeq \frac{1}{d_{\text{eff}}}. \tag{10}$$

Thus, conditional on the active amplitudes,

$$\mathrm{Var}\left(\sum_{\substack{j \in S \\ j \neq i}} x_j \, w_i^{\mathsf{T}} w_j \right) \simeq \frac{1}{d_{\text{eff}}} \sum_{\substack{j \in S \\ j \neq i}} x_j^2. \tag{11}$$

For amplitudes of order unity,

$$\sigma_{\text{int}} \sim \sqrt{\frac{k}{d_{\text{eff}}}}. \tag{12}$$

This is the readout bottleneck. Concentration lets many directions coexist because their overlaps are small. But readout sees the *sum* of many small overlaps. Reliable recovery requires

$$\sqrt{\frac{k}{d_{\text{eff}}}} \ll \Delta, \tag{13}$$

where Δ is the relevant decision margin. Equivalently,

$$\text{kinetic: } N \lesssim \exp(c \, d_{\text{eff}} \, \varepsilon^2), \qquad \text{epistemic: } k \lesssim d_{\text{eff}} \, \Delta^2. \tag{14}$$

The first is storage-like coexistence. The second is simultaneous accessibility. A model may host many more features than it can jointly read.

This also clarifies the relation to compressed sensing. Stable recovery of a k-sparse vector in an N-dimensional feature dictionary from m measurements typically requires $m \gtrsim Ck \log(N/k)$ under suitable incoherence or restricted-isometry assumptions [10–12]. In an idealized isotropic residual stream, one may identify m with d_{eff}; in trained models the effective representation dimension and effective readout dimension may differ. The compressed-sensing analogy is therefore diagnostic, not literal.

III. HINGE VECTORS AND SENSE SUBSPACES

The previous section established an architecture-independent capacity constraint. We now ask a more speculative question: how might a trained model organize readout within that constraint? The hinge-and-frame formalism is one possible answer. It is not implied by concentration geometry, and it is separately falsifiable.

A. The hinge hypothesis

For a token t, let $v(t) \in \mathbb{R}^d$ be a candidate unit vector, called a *hinge*. Empirically this could be the static input embedding, a mean contextual embedding, or a learned token-specific probe direction. The hypothesis is not that every vector decomposition is meaningful. Any state can be decomposed parallel and perpendicular to any direction. The claim is stronger: for some tokens, a token-associated direction is privileged relative to matched controls.

Given a context C, the hinge is the shared first vector of a sense frame,

$$e_1^C = v(t). \tag{15}$$

The carrier for sense information is the orthogonal complement $v(t)^{\perp}$. A context selects a low-dimensional sense subspace

$$S^C = \mathrm{span}\{e_2^C, \ldots, e_{q_C+1}^C\} \subset v(t)^{\perp}, \tag{16}$$

with $q_C \ll d_{\text{eff}}$ in the intended regime. For example, $v(\text{bank})$ might be shared by finance, river, and aircraft-tilt sense subspaces.

The low-rank assumption is empirical. A word sense is not a single direction: it may involve topic, syntax, selectional preferences, and world knowledge. But it should not need the full residual stream either. The hinge hypothesis predicts that dominant within-sense variation is low-dimensional after projection to the hinge-perpendicular carrier. This prediction is compatible with existing evidence that contextualized embeddings can separate word senses, but it is stronger: it asks whether such separation is privileged relative to a token-associated hinge [13, 14].

B. Gated-subspace readout

Let $h(x) \in \mathbb{R}^d$ be the contextual state for input context x. Relative to the hinge,

$$\begin{aligned} h(x) &= h_{\parallel}(x) v(t) + h_{\perp}(x), \\ h_{\parallel}(x) &= v(t)^{\mathsf{T}} h(x), \\ h_{\perp}(x) &\in v(t)^{\perp}. \end{aligned} \tag{17}$$

The scalar $h_\|$ measures engagement with the token-associated direction. The perpendicular component $h_\perp$ carries the sense-disambiguating information.

Readout inside S^C requires conditioning. Pairwise small overlaps are not enough. Let

$$F_S^C = [\, e_2^C \;\; \cdots \;\; e_{q_C+1}^C \,], \qquad G_S^C = (F_S^C)^\mathsf{T} F_S^C. \tag{18}$$

We call the sense frame ρ-stable if

$$\|G_S^C - I\|_{\mathrm{op}} \le \rho < 1. \tag{19}$$

Then the dual frame

$$\widetilde{F}_S^C = F_S^C (G_S^C)^{-1} \tag{20}$$

is well-defined and satisfies $\|\widetilde{F}_S^C\|_{\mathrm{op}} \le (1-\rho)^{-1/2}$. Near-degenerate frames amplify readout noise; the model requires ρ comfortably below one.

The dual-frame coefficients are

$$\alpha^C(x) = \big(\widetilde{F}_S^C\big)^\mathsf{T} h_\perp(x) \in \mathbb{R}^{q_C}. \tag{21}$$

A simple sense score is

$$s_C(x) = g(h_\|(x))\, r_C(h_\perp(x)), \tag{22}$$

where $g \ge 0$ is an engagement gate and $r_C \ge 0$ measures how well $h_\perp(x)$ fits the sense subspace. One may take $r_C = \|\alpha^C(x)\|^2$, a squared overlap with a distinguished direction in S^C, or a learned positive quadratic form. The selected sense frame is

$$C^*(x) = \arg\max_C s_C(x). \tag{23}$$

The precise gate is not essential. The empirical content is the separation: the parallel coordinate gates token engagement; the perpendicular carrier contains sense information.

The hinge matters because different candidate hinges induce different carriers. If no token-associated hinge exposes sense structure better than matched random or frequency-controlled directions, the hinge hypothesis fails.

C. Packing sense subspaces around one hinge

The vector packing bound extends to low-dimensional subspaces. Fix a unit vector $v_0 \in S^{d-1}$ and let $m = d-1$. For $1 \le q \ll m$ and any fixed threshold $\mu > 2\sqrt{q/m}$, there exist q-dimensional subspaces $S_1, \dots, S_M \subset v_0^\perp$ such that

$$\|U_a^\mathsf{T} U_b\|_{\mathrm{op}} \le \mu, \qquad a \ne b, \tag{24}$$

where $U_a, U_b \in \mathbb{R}^{m\times q}$ are orthonormal basis matrices, and

$$M \ge \exp(c_{\mu,q} m) \tag{25}$$

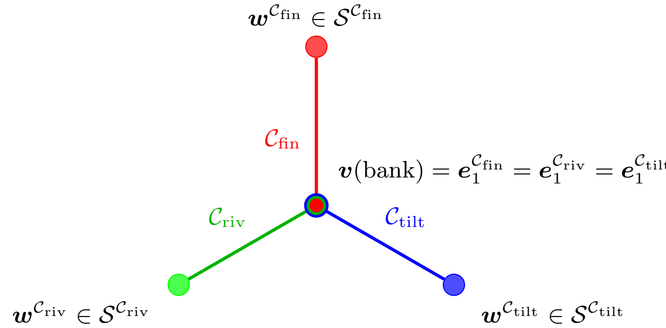

FIG. 1. Schematic of hinge-anchored frame sharing. A candidate hinge vector associated with the token "bank" is shared by several possible sense frames. The colored outer nodes represent representative sense directions, not entire subspaces. The formal model treats sense content as lying in low-dimensional subspaces of the hinge-perpendicular carrier $\boldsymbol{v}(\mathrm{bank})^\perp$.

for some positive $c_{\mu,q}$.

This is a standard Grassmannian packing consequence of concentration on Stiefel and Grassmann manifolds [15, 16]. For two independent Haar-random q-planes, the singular values of $U^\mathsf{T} V$ are the cosines of their principal angles. In the regime $q \ll m$, the largest principal correlation is of order $\sqrt{q/m}$ [17–19]. A union bound over sampled subspaces then gives an exponential packing.

For trained representations this statement must be read inside an approximately isotropic effective carrier of dimension $m_{\mathrm{eff}} = d_{\mathrm{eff}} - 1$, not necessarily inside the raw ambient space. Without such an effective carrier, the Haar-random subspace model is not a good description of activation geometry.

D. Relation to dynamic contextual embeddings

Transformers already handle polysemy dynamically. The state $h_\ell(t \mid x)$ of token t at layer ℓ depends on context x. The hinge formalism is not an alternative architecture; it is a possible decomposition of such trajectories.

Given a candidate hinge h_t, write

$$\begin{aligned} h_\ell(t \mid x) &= a_\ell(t,x) h_t + z_\ell(t,x), \\ a_\ell(t,x) &= h_t^\mathsf{T} h_\ell(t \mid x), \\ z_\ell(t,x) &\in h_t^\perp. \end{aligned} \tag{26}$$

The decomposition is trivial; the hypothesis is not. Hinge anchoring predicts that a_ℓ tracks token engagement or salience, while z_ℓ carries most sense-disambiguating information. Sense labels should become more separable in $h_t^\perp$ than in the orthogonal complements of matched control directions. For each sense s, the vectors $z_\ell(t,x)$ should also concentrate near a low-dimensional subspace $S_s \subset h_t^\perp$.

Thus the critical question is not whether contextual embeddings move; they do. The question is whether their movement has a privileged hinge-perpendicular organization.

IV. ATTENTION AND LATENT FRAME SELECTION

A standard attention head computes

$$h' = \sum_j \alpha_j \nu_j, \qquad \alpha_j = \frac{\exp(q^\top k_j/\sqrt{d})}{\sum_{j'} \exp(q^\top k_{j'}/\sqrt{d})}. \tag{27}$$

Keys and values are produced by different learned maps, so values are not generally dual-frame coefficients for keys. Attention is therefore not literal dual-frame reconstruction.

The useful analogy is softer. Attention routes information: it amplifies some contextual directions and suppresses others. In the hinge language, this resembles subspace or frame selection. A literal dual-frame interpretation would require additional constraints, such as approximately Parseval keys and values implementing the corresponding dual reconstruction. We do not assume this.

A compact latent-variable version is

$$p(t \mid x) = \sum_C p(t \mid x, C)\, p(C \mid x), \tag{28}$$

where C indexes candidate contextual frames. One may model

$$p(C \mid x) = \frac{\exp(s_C(x)/T)}{\sum_{C'} \exp(s_{C'}(x)/T)}, \tag{29}$$

with s_C the gated score of Eq. (22). This is not a claim about the literal implementation of decoding in current LLMs: standard temperature acts on token logits, not on explicit latent frames. The point is only that ambiguity can be represented as a distribution over possible readout frames rather than as a single fixed interpretation.

V. DISCUSSION

The main proposal is the capacity/accessibility distinction. The hinge formalism is a possible organization within that constraint, not a consequence of it.

A. Relation to superposition and sparse autoencoders

The superposition hypothesis holds that neural networks represent more features than they have neurons by packing features into nearly orthogonal directions [5, 20]. The concentration estimate (5) gives the kinetic side of that picture: many directions can coexist. Equation (12) gives the epistemic side: many jointly active features create interference of scale $\sqrt{k/d_{\text{eff}}}$. Sparsity is therefore not an implementation detail; it is necessary for readout.

Sparse autoencoders (SAEs) give a concrete empirical interface. They approximate layer activations as sparse expansions over learned decoder directions, making the w_i of Eq. (8) measurable objects rather than hypothetical features. Recent work has scaled SAEs, studied SAE feature geometry, and automated interpretation of large numbers of latents [21–24]. In the present framework, SAE decoder directions can test the capacity/accessibility prediction, while sense-conditioned SAE activation clusters can test the stronger hinge hypothesis.

B. A coactivation-weighted capacity limit

The active-set size k is a crude summary. A more refined description uses the coactivation graph

$$C_{ij} = \mathbb{P}(i, j \text{ active}). \tag{30}$$

If active amplitudes have, conditional on coactivation, zero mean and pairwise uncorrelated signs, then cross terms cancel in expectation and feature j contributes to the readout variance of feature i in proportion to $\mathbb{E}[x_j^2 \mid i, j \text{ active}](w_i^\top w_j)^2$. Absorbing amplitude variance and feature importance into coefficients a_i, a_j gives

$$E_{\text{int}} = \sum_{i \neq j} C_{ij} a_i a_j (w_i^\top w_j)^2. \tag{31}$$

Frequently coactive features are therefore pressured toward orthogonality. Rarely coactive or mutually exclusive features are much less constrained and may share directions or even become approximately antipodal if nonlinear readout makes this useful. The relevant object is not just the number of features, but the weighted coactivation graph.

C. Limitations

The framework can fail in several ways.

First, anisotropy may make d_{eff} much smaller than d. Contextual embeddings are known to be anisotropic, and intrinsic dimension can vary strongly by layer and training regime [25–28]. Whitening and related postprocessing can partially restore isotropic geometry [26, 27, 29]. At the same time, anisotropy should not be treated as an automatic or universal property of all Transformer representations [30]. The bounds should always be read in terms of the effective dimension actually achieved.

Second, the hinge readout needs well-conditioned sense frames. If ρ in Eq. (19) is close to one, the dual-frame readout amplifies noise.

Third, the hinge pattern may simply be descriptively wrong. Trained models may represent polysemy without

TABLE I. Dynamic contextual embeddings versus the hinge-decomposition hypothesis.

Dynamic description	Hinge-decomposition hypothesis
Token state changes by context and layer	State decomposes relative to a hinge
Layer activation carries context	Hinge-perpendicular part carries sense
Polysemy by vector deformation	Polysemy by gated subspace readout
Resource is depth and width	Resource is quasi-dimensionality
Limit is activation bottleneck	Limit is interference/accessibility

any preferred token-associated direction. In that case the capacity/accessibility account may still hold, but the hinge hypothesis fails.

Fourth, the law $\sigma_{\rm int} \sim \sqrt{k/d_{\rm eff}}$ assumes random-like incoherent directions and weakly correlated signs or amplitudes. Learned networks may improve on this by orthogonalizing frequently coactive features, or worsen it through anisotropy and correlated errors. The law is a scaling principle, not a universal equality.

Finally, the hinge formalism has modeling choices: the hinge, the gate, the selector, and the sense-subspace dimension. These choices must be fixed or controlled empirically. Otherwise the framework risks describing whatever geometry is found after the fact.

D. Empirical predictions

The key empirical task is to separate the general capacity claim from the stronger hinge claim.

a. Claim A: hinge anchoring. For a polysemous token t, there exists a privileged token-associated direction h_t such that sense-relevant variation is better exposed in $h_t^\perp$ than in the orthogonal complements of matched control directions.

b. Claim B: low-dimensional sense frames. Within $h_t^\perp$, sense-conditioned states concentrate near low-dimensional subspaces with small cross-sense coherence.

These claims can fail independently. Claim A can hold while sense information remains high-dimensional. Claim B can hold relative to some non-token-associated direction, in which case hinge anchoring fails.

The following tests separate the possibilities.

1. *Hinge-anchoring test.* For a polysemous token, choose candidate hinges such as the static input embedding, mean contextual embedding, or a learned token-specific direction. Decompose layer states as in Eq. (26). Sense labels should be better separated in $h_t^\perp$ than in orthogonal complements of random, frequency-matched, or layer-matched control directions. The statistic may be classification accuracy or mutual information with sense labels, assessed by permutation over sense labels and bootstrap over token instances.

2. *Operational subspace test.* For each sense s, collect projected states $z_\ell(t,x) \in h_t^\perp$ and estimate a low-rank subspace by PCA or probabilistic factor analysis. With U_s the first q principal directions, measure cross-sense coherence $\|U_s^{\mathsf T} U_{s'}\|_{\rm op}$. True sense labels should yield lower cross-sense coherence than shuffled labels and stronger separation than control hinges.

3. *Gated-readout test.* Linear probes trained on $z_\ell(t,x)$ should perform comparably to probes trained on the full state $h_\ell(t \mid x)$, while probes restricted to the parallel component $a_\ell(t,x)h_t$ should perform near chance on sense classification.

4. *Overlap-distribution test.* If a representation exploits quasi-dimensionality, pairwise inner products among token embeddings or SAE feature directions should concentrate near zero with variance $\sim 1/d_{\rm eff}$. Deviations quantify anisotropy and effective dimension loss.

5. *Synthetic concurrency degradation.* At a fixed layer, inject controlled superpositions
$$h_k = h_0 + \lambda \sum_{j=1}^{k} x_j w_j, \tag{32}$$
using directions w_j obtained independently, for example from SAE decoder directions or frozen linear probes. As k increases, readout should degrade smoothly on the scale $\sqrt{k/d_{\rm eff}}$ for random-like directions, with weaker degradation for feature sets that training has orthogonalized.

6. *Coactivation-dependent geometry.* Frequently coactive features should show reduced mutual coherence relative to rarely coactive features. This tests the broader capacity/accessibility framework through Eq. (31), independently of the hinge hypothesis.

E. Scope of the quantum analogy

The analogy to quantum contextuality is structural and limited. Both settings involve vectors whose operational role depends on a frame. But quantum contextuality has the force of no-go theorems such as Kochen–Specker [31, 32]; no analogous theorem is claimed here for semantic embeddings. The borrowed idea is only the shared-vector frame structure: one vector can participate in several possible frames, and readout depends on which frame is selected.

VI. CONCLUSION

High-dimensional representations can host many semantic possibilities, but finite readout limits how many can be made simultaneously accessible. The main quantitative content is the same pair of scalings introduced in Eqs. (1) and (2):

$$N \lesssim \exp(c\, d_{\mathrm{eff}}\, \varepsilon^2), \qquad \sigma_{\mathrm{int}} \sim \sqrt{k/d_{\mathrm{eff}}}. \tag{33}$$

The first expression is a coexistence bound; the second is a readout noise scale. They are not the same kind of theorem, but together they describe the difference between semantic capacity and semantic accessibility.

Within this framework we proposed the hinge-and-frame hypothesis: some polysemous tokens may be organized around stable token-associated directions, with sense information encoded in low-dimensional subspaces of the hinge-perpendicular carrier. This hypothesis is not forced by geometry. Its critical test is whether token-associated hinges expose sense structure better than matched controls. A negative result would falsify hinge anchoring while leaving the broader superposition and interference picture intact.

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